

Hundsdoerfer + Verwer I.7

We say the ODE system

$$u'(t) = f(u, t)$$

is non-negativity preserving ("positive") if

$$u(0) \geq 0 \Rightarrow u(t) \geq 0 \quad \forall t > 0.$$

For example u may represent:

- Density, pressure
- Concentration
- Probability
- Water depth

Linear Problems

$$u'(t) = Au(t)$$

$$u(0) = u_0 \geq 0$$

What are the conditions on A so that $u(t) \geq 0 \quad \forall t > 0$?

Positivity of matrix-vector product

What are the conditions so that

$$x \geq 0 \Rightarrow Ax \geq 0?$$

$$a_{ij} \geq 0 \quad \forall i, j$$

$u(t) = e^{tA} u(0)$ so we need $e^{tA} \geq 0$.

$e^{hA} = I + hA + O(h^2) \Rightarrow I + hA$ must be ≥ 0 .
(for small enough h)

$\Rightarrow a_{ij} \geq 0$ for $i \neq j$. (1) (necessary)

Sufficiency:

$e^{tA} = \lim_{\substack{h \rightarrow 0 \\ N = t/h}} (I + hA)^N \leftarrow \text{for every } h, \text{ this is positive if (1) holds.}$
 so this is also positive. $h = t/N$

$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n$

Nonlinear case

$u'(t) = f(u, t)$

$u(0) \geq 0$

Thm: Assume there exists a unique solution. Then the system is positive iff for all u, i

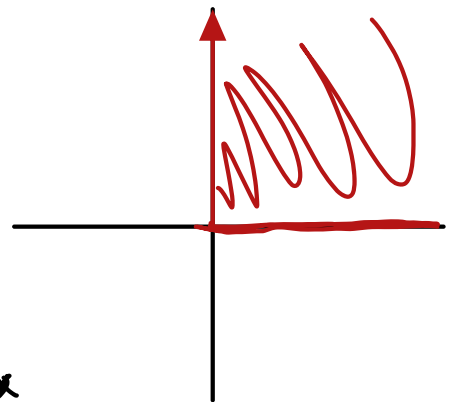
$u \geq 0, u_i = 0 \Rightarrow f_i(u, t) \geq 0$.

PDE semidiscretizations

Heat eqn., Centered Diff.: $u_t = u_{xx}$

$u'_i(t) = \frac{1}{\Delta x^2} (u_{i+1} - 2u_i + u_{i-1})$ Is it positive?

$u'(t) = Au$ $a_{ij} \geq 0 \forall i \neq j \Rightarrow \text{Yes.}$



Advection: $u_t = u_x$

Semi-discretize:

- Upwind $u'_i(t) = \frac{1}{\Delta x} (u_{i+1} - u_i)$
 $u'(t) = Au$ $A = \frac{1}{\Delta x} \begin{bmatrix} -1 & +1 & & \\ & -1 & +1 & \\ & & \ddots & \ddots \\ & & & -1 & +1 \\ & & & & -1 \end{bmatrix}$ ✓

- Centered:

$$u'_i(t) = \frac{1}{2\Delta x} (u_{i+1} - u_{i-1}) \quad \text{Not positive.}$$

Advection-diffusion: $u_t = au_x + bu_{xx}$ $b \geq 0$

Centered diff.:

$$u'_i(t) = a \frac{u_{i+1} - u_{i-1}}{2\Delta x} + b \frac{u_{i+1} - 2u_i + u_{i-1}}{(\Delta x)^2}$$

$$= \frac{-b}{(\Delta x)^2} u_i + \left(\frac{a}{2\Delta x} + \frac{b}{(\Delta x)^2} \right) u_{i+1} + \left(\frac{-a}{2\Delta x} + \frac{b}{(\Delta x)^2} \right) u_{i-1}$$

must be ≥ 0

$$\frac{b}{(\Delta x)^2} \pm \frac{a}{2\Delta x} \geq 0 \quad \frac{b}{\Delta x} \pm \frac{a}{2} \geq 0 \quad \left| \frac{a}{2} \right| \leq \frac{b}{\Delta x}$$

$$\Delta x \leq \frac{2b}{|a|}$$

Positivity of ODE discretizations

Given a positive ODE

$$u'(t) = Au(t) \quad a_{ij} \geq 0 \quad i \neq j$$

will our numerical solution remain positive?

Explicit Euler

$$u^{n+1} = (I + hA)u^n$$

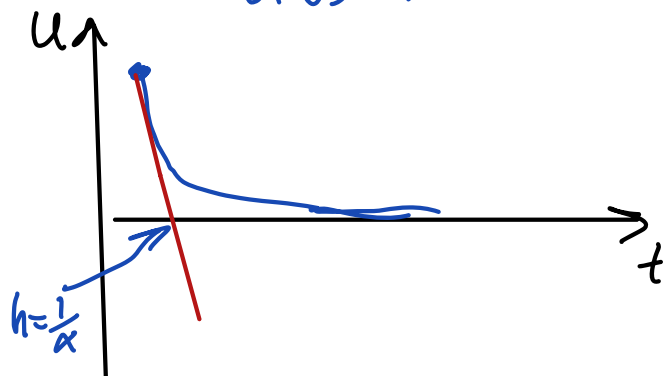
So we need $I + hA \geq 0$. Let $-\alpha = \min_i a_{ii} < 0$

Then we need $1 - h\alpha \geq 0 \Leftrightarrow h \leq \frac{1}{\alpha}$.

In the scalar case, abs. stab. requires $h \leq \frac{2}{\alpha}$.

Positivity is more exacting.

$$u'(t) = \lambda u \quad \lambda < 0$$



Is abs. stability ever more restrictive than positivity for positive, linear ODEs? (I don't know)

Backward Euler

$$u^{n+1} = (I - hA)^{-1}u^n$$

we need this to exist

If $I - hA$ is singular, then $(I - hA)x = 0 \Leftrightarrow x = hAx$

so A has an eigenvalue $\frac{1}{h} > 0$.

$$\frac{1}{h}x = Ax$$

So we assume

A has no positive real eigenvalues.

Then do we have

$$(I - hA)^{-1} \geq 0?$$

Dfn. We say $M \in \mathbb{R}^{n \times n}$ is an M-matrix
if M is invertible and $M^{-1} \geq 0$.

Thm. Let $m_{ij} \leq 0 \ \forall \ i \neq j$. Then M is an M-matrix
iff every real eigenvalue of M is positive.

If λ is an eigenvalue of A , the corresponding
eigenvalue of $I - hA$ is $1 - h\lambda$.

$$\lambda < 0 \Rightarrow 1 - h\lambda > 0.$$

So $I - hA$ is an M-matrix, so $(I - hA)^{-1} \geq 0$.

So Backward Euler is positive (unconditionally).

This is even true for any positive nonlinear ODE
system.

Unfortunately, no higher order method is unconditionally
positive for all positive linear ODEs.

Thm. (Bolley + Crouzeix 1978)

Let a general linear method be given with order of accuracy $p > 1$. Then there exists a positive linear ODE

$$u'(t) = Au \quad a_{ij} \geq 0 \quad i \neq j$$

$$u(0) = u_0 \geq 0$$

and a step size $0 < h < \infty$ such that the method yields $u_i \neq 0$.